EXERCISE 1

Consider the function f defined by

$$f(x) = xe^{3\cos x}, \quad \text{for } -\pi \leq x \leq \pi.$$

The point A is the point on the graph of f for which the x-coordinate is $\frac{\pi}{2}$.

1. Show algebraically that the gradient of the tangent to the graph of f at point A is $\frac{2-3\pi}{2}$. [4 marks]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

2. Hence find the equation of the normal line to the graph of f at point A, and determine where that line intersects the x-axis. [5 marks]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

EXERCISE 2 Find the equations of the horizontal tangents to $y = 2x^3 + 3x^2 - 12x + 1$. [5 marks]